Department of CSE(AIML), PES University
UE26CS151A -Python for Computational Problem Solving
Laboratory Week

Note: Students are required to write the programs along with atleast two outputs for each experiment in the laboratory observation record. The observation record must be signed by the faculty in charge. For every laboratory session, the observation component carries **3 marks**.

1) Write a Python program to accept the marks of a student in **5 subjects**. Calculate the total marks, average, and percentage using arithmetic operators. Display the grade based on the following conditions:

- 90–100: Grade A
- 75–89: Grade B
- 60–74: Grade C
- 50–59: Grade D

2) A student is eligible for a scholarship if:

- Marks are **80 or above**, and
- Family income is **less than ₹3,00,000**.

  Write a Python program to check the eligibility

3) Write a Python program to simulate an ATM withdrawal. Accept the account balance and withdrawal amount. Check the following conditions:

- Withdrawal amount must be positive.
- Withdrawal amount must be a multiple of ₹100.
- Withdrawal amount must not exceed the available balance.

4) Write a Python program to accept a three-digit number and calculate:

- Sum of its digits
- Product of its digits
- Reverse of the number
- Whether the number is a palindrome
- Whether the sum of its digits is divisible by 3

5)A customer purchases 5 items. Write a Python program to accept the price of each item using a loop and calculate the total bill.

6) Write a Python program to accept the temperature of a city and display:

- **"Cold"** if temperature is below 20°C
- **"Moderate"** if temperature is between 20°C and 30°C
- **"Hot"** if temperature is above 30°C.

7) Write a Python program to accept a person's age and calculate the movie ticket price:

- Below 12 years → ₹100
- 12–59 years → ₹200
- 60 years and above → ₹120

8) Write a Python program to accept the expenses for **7 days** using a loop and calculate the **total weekly expense**.

9)Write a Python program to accept the amount of water consumed by a person for **7 days** and calculate the average daily water consumption.

10)A parking area charges:

- First 2 hours → ₹30
- 3–5 hours → ₹50
- More than 5 hours → ₹100

Write a Python program to accept the number of hours a vehicle is parked and calculate the parking fee.

Prepared By,
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